

DHANASHALINI S

Software Engineer

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Software Engineer building backend and applied-AI systems in Python, TypeScript and Node.js — distributed job processing, retrieval-augmented generation, and LLM-assisted developer tooling. Breaks complex problems into maintainable, tested solutions and decides from data: benchmarked retrieval, regression gates in CI, and 25–28% gains on production ML pipelines. Analytical, detail-oriented and quick to pick up unfamiliar technologies; owns work end to end, documents trade-offs, and improves through code review.

EXPERIENCE

Truerize IQ Strategic Solutions Pvt. Ltd. — Software Engineer Intern

March 2026 – May 2026

- Raised land-cover classification accuracy **28%** on satellite imagery by building end-to-end CNN and U-Net semantic-segmentation pipelines and deploying them as FastAPI REST services with SHAP and LIME explainability.
- Cut batch and real-time inference latency **25%** by vectorising HSV, LBP and HOG feature extraction over Polars and PyArrow columnar pipelines, replacing per-image Python loops.

PROJECTS

CodeLens — AI Code Review & Repository Intelligence Platform [github.com/dhanashalini25/codelens]

Python · FastAPI · GitHub Actions · NumPy · SQLite · LLM APIs

- Cut pull-request review turnaround to **under 90 seconds** by shipping a GitHub Action running **13** AST-based static rules before the LLM pass, blocking merges on committed credentials and shell-injection risks; **48** tests green on Python 3.10–3.12.
- Improved code-search top-1 accuracy **40%** — from 5 to 7 of 10 benchmark queries — by weighting symbol names in a pure-NumPy hashed TF-IDF index, and rejected BM25 length normalisation after benchmarking it 2 points worse.

Ibid — Grounded Document Question Answering (RAG) [github.com/dhanashalini25/ibid]

Python · FastAPI · Retrieval-Augmented Generation · Embeddings · SQLite

- Achieved **100%** refusal accuracy on unanswerable questions by gating answers behind a tuned retrieval-score threshold plus a model-side NO_ANSWER contract, eliminating unsupported claims.
- Lifted recall@5 from **95% to 100%** (MRR 0.950 to 0.967) via keyword-overlap reranking, measured by a 26-question harness that gates CI on recall and refusal floors to block silent regressions.

PulseQueue — Distributed Background Job Processing Platform [github.com/dhanashalini25/pulse-queue]

Node.js · Express · BullMQ · Redis · Docker · REST APIs

- Sustained job delivery through worker crashes via exponential-backoff retries across **4** priority tiers, with a dead-letter queue isolating exhausted jobs for replay instead of dropping them.
- Reduced failure-triage time **60%** by surfacing queue depth, worker activity and throughput through a monitoring dashboard, while scaling API and worker tiers independently against shared Redis via Docker Compose.

DevHire — Full-Stack Developer Hiring Platform [github.com/dhanashalini25/devhire-ai-hiring-platform]

Next.js · TypeScript · Node.js · Express · Prisma · PostgreSQL · JWT

- Enabled **2** distinct role workflows for candidates and recruiters via JWT authentication with role-based authorization across protected Next.js routes and Express APIs.
- Eliminated **100%** of duplicate applications with a PostgreSQL uniqueness constraint enforced in Prisma, and cut list-endpoint payloads **70%** by moving search, filtering and pagination server-side.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, SQL

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, Software Design, API Design, Debugging, Unit and Integration Testing

Backend: Node.js, Express, FastAPI, REST APIs, Prisma, BullMQ, async workers, concurrency, rate limiting, JWT authentication

Frontend: React, Next.js, Tailwind CSS, HTML5, CSS3, Responsive Design

Data & ML: RAG, embeddings, vector search, reranking, LLM APIs, prompt engineering, PyTorch, TensorFlow, scikit-learn, Pandas, Polars, NumPy, feature engineering, Explainable AI (XAI)

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Redis, database design, SQL query optimization, data modeling

Cloud & Practices: Docker, Docker Compose, GitHub Actions, CI/CD, AWS EC2, AWS S3, Git, Linux, Postman, code review, performance optimization

EDUCATION & CERTIFICATIONS

B.Tech, Information Technology — Knowledge Institute of Technology, 2025 | CGPA 7.8/10

Google Cybersecurity Professional Certificate | Google Data Analytics Professional Certificate — Coursera